NYC GREEN TAXI · Q1 2025

Daily revenue, January to March 2025

$3.40M	145,743	90	$23.34
total revenue	paid trips	days	average per trip

Data pipeline project · Source: NYC TLC Green Taxi trip records, Jan–Mar 2025

HOW IT WORKS

Five stages turn raw trips into a daily table

INGEST

Download

3 monthly Parquet files, retried up to 3 times

→

EXTRACT

Read

2 of 21 columns, 146,486 rows

→

TRANSFORM

Clean

Drop trips outside Q1 and amounts of $0 or less

→

TRANSFORM

Aggregate

Sum total_amount per pickup day

→

LOAD

Write

90 rows as CSV and Parquet

ETL, not ELT

The data is small and there’s no database, so pandas cleans it before only the final table is saved.

Orchestrated with Prefect

Each stage is a task in one flow. Only the download retries, because it’s the only step that uses the network.

Code: src/taxi_pipeline/ · download.py, extract.py, transform.py, load.py, flow.py

DATA QUALITY

Cleaning kept 99.5% of trips

RAW

146,486

rows in 3 monthly files

→

IN PERIOD

146,475

−11 picked up outside Jan 1 – Mar 31

→

CLEAN

145,743

−732 with total_amount of $0 or less

Why drop out-of-period trips?

Pickups from 2024-12-25 to 2025-04-01 would add stray days to a Q1 report.

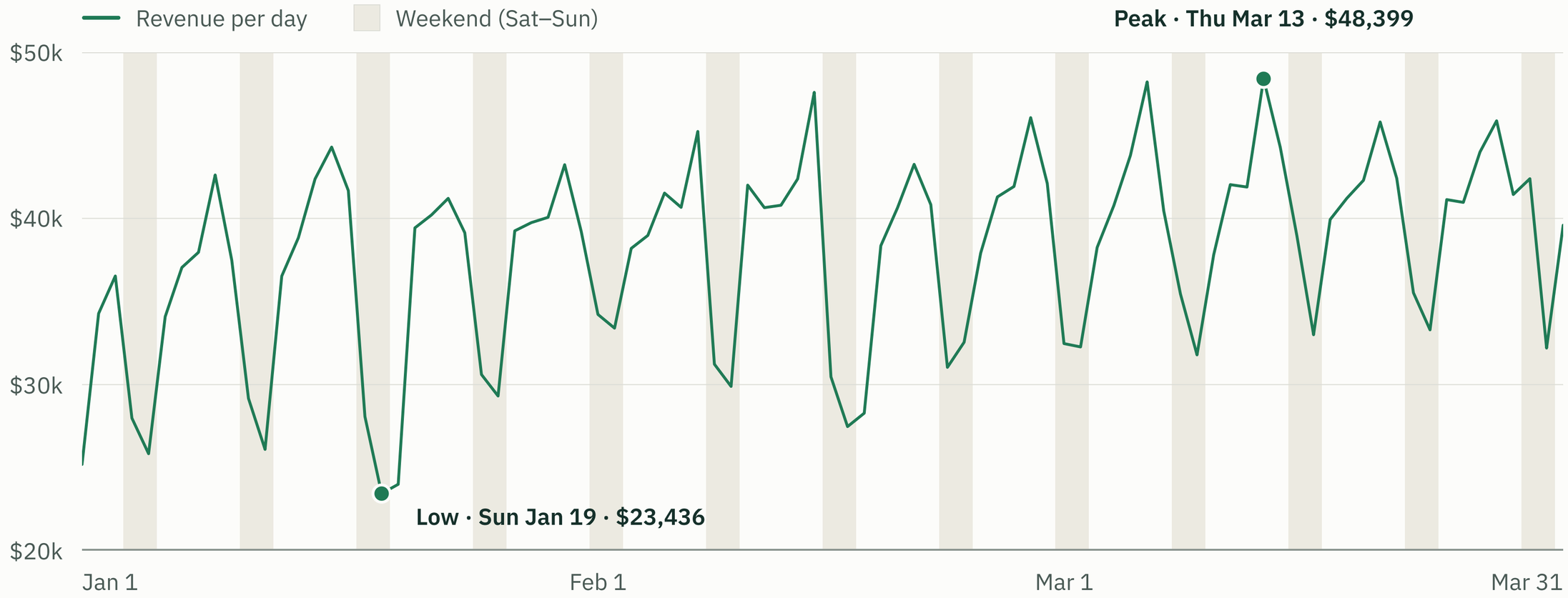
Why drop amounts of $0 or less?

478 negative rows are mostly “No charge” (337) and “Dispute” (127) reversals; 254 rows are exactly $0.

Rules decided in notebooks/explore.ipynb · applied by clean_trips() in transform.py

REVENUE PER DAY

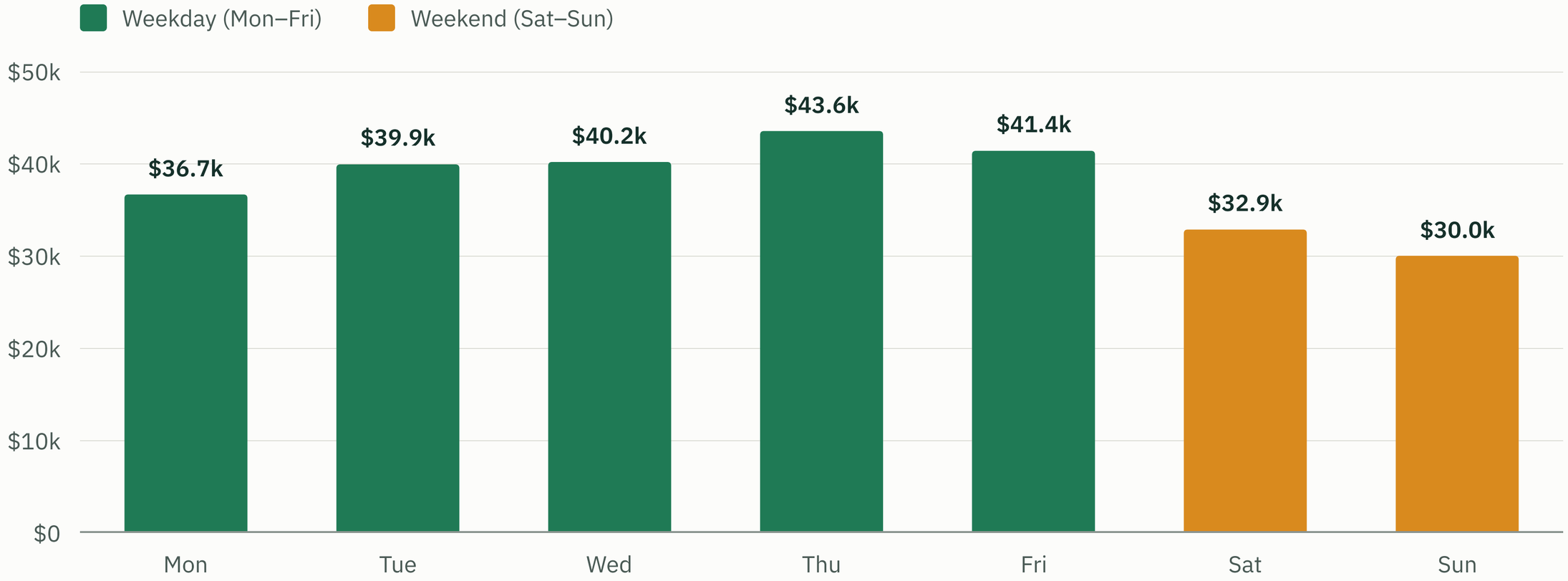
Revenue dips every weekend



Source: data/processed/daily_revenue.csv · revenue = total_amount · y-axis starts at $20k · average day $37,797

AVERAGE REVENUE BY WEEKDAY

Thursdays earn the most



Average of each weekday’s daily revenue, Jan 1 – Mar 31, 2025 (12–13 days per weekday)

MONTH BY MONTH

March was the strongest month

January

$35,319

average revenue per day

$1,094,901 total

48,103 trips · 31 days · $22.76 per trip

February

$38,176

average revenue per day

$1,068,929 total

46,405 trips · 28 days · $23.03 per trip

March

$39,933

average revenue per day

$1,237,923 total

51,235 trips · 31 days · $24.16 per trip

Revenue per day grew **13%** from January to March, and so did the average fare per trip. February's total is lowest only because it has 28 days.

Bars show each month's revenue per day relative to March · totals rounded to the dollar

BEST AND WORST DAYS

The best days were Thursdays

Top 5 days

Date	Day	Trips	Revenue
Mar 13	Thursday	1,995	$48,399
Mar 6	Thursday	2,008	$48,219
Feb 14	Friday	2,081	$47,588
Feb 27	Thursday	1,948	$46,066
Mar 27	Thursday	1,847	$45,875

Bottom 5 days

Date	Day	Trips	Revenue
Jan 19	Sunday	1,105	$23,436
Jan 20	Monday	1,081	$23,997
Jan 1	Wednesday	998	$25,195
Jan 5	Sunday	1,127	$25,838
Jan 12	Sunday	1,140	$26,096

All five quietest days are in January: three Sundays, New Year’s Day (Jan 1), and Martin Luther King Jr. Day (Jan 20). The one non-Thursday at the top is Valentine’s Day.

Source: data/processed/daily_revenue.csv · revenue rounded to the dollar

TAKEAWAYS

What the result shows

A weekly rhythm

An average Thursday ($43.6k) earns about 45% more than an average Sunday ($30.0k).

Growth over the quarter

Revenue per day rose 13%, from $35.3k in January to $39.9k in March.

Holidays are quiet

New Year’s Day and Martin Luther King Jr. weekend were among the five lowest days.

Known limitation

Dropping negative rows keeps the charges they reverse, so revenue is slightly overstated.

Run it yourself: uv run scripts/download_data.py, then uv run scripts/run_pipeline.py